



C S C I 8314

Spring 2023

SPARSE MATRIX COMPUTATIONS

***Class time* : MW 9:45 – 11:00am**
***Room* : Ackerman Hall 211**
***Instructor* : Yousef Saad**

August 7, 2026

About this class: Objectives

Set 1 An introduction to sparse matrices and sparse matrix computations.

- Sparse matrices;
- Sparse matrix direct methods ;
- Graph theory viewpoint; graph theory methods;

Set 2 Iterative methods and eigenvalue problems

- Iterative methods for linear systems
- Algorithms for sparse eigenvalue problems and the SVD
- Possibly: nonlinear equations

Set 3 Applications of sparse matrix techniques

- Applications of graphs; Graph Laplaceans; Networks ...;
- Standard Applications (PDEs, ..)
- Applications in machine learning
- Data-related applications
- Other instances of sparse matrix techniques

➤ Please fill out (now if you can)

This survey

Short link url:

<https://forms.gle/i5MCBg3X289JMAHd8>

Who is in this class today?

- Out of 20 [as of Tuesday] - registered
 - 5 in Computer Science
 - 5 in Aerospace Engineering
 - 2 Electrical Engineering
 - 2 Civil Engineering
 - 2 Chemical Engineering/ Materials Science
 - 2 Mathematics
 - 1 Statistics
 - 1 Industrial & Systems Eng.

Logistics:

- Lecture notes and minimal information will be located here:
[8314 at cselabs class web-sites](#)

URL:

<https://www-users.cselabs.umn.edu/classes/Spring-2023/csci8314>

[also follow: 'teaching' at www.cs.umn.edu: /saad]

- There you will find :
 - Lecture notes, Schedule of assignments/ tests, class info
- Canvas will contain the rest of the information: assignments, grades, etc.

About lecture notes:

- Lecture notes (like this first set) will be posted on the class web-site – usually before the lecture.
- Review them to get some understanding if possible before class.
- Read the relevant section (s) in the texts or provided references
- Lecture note sets are grouped by topics rather than by lecture.
- In the notes the symbol  indicates suggested easy exercises or questions – often [not always] done in class.
- Also: occasional practice exercises posted

Matlab, Python-Numpy, etc..

- Important to use either Matlab (mostly) or Python to quickly illustrate and test algorithms.
- Scripts in either matlab or python will be posted in the 'matlab' section of the class web-site.
- Also: matlab or python demos seen in class will be posted

Roadmap – [subject to itinerary change!]

Part 1

1. Sparse matrices;
2. Graph representations;
3. Sparse direct methods for linear systems;

Part 2

4. Iterative methods for linear systems ;
5. Projection methods and Krylov subspace methods;
6. Eigenvalue problems;

Part 3

7. Back to Graphs; Paths in graphs; Markov Chains;
8. Graph centrality;
9. Graph Laplaceans and applications; Clustering;
10. Graph embeddings.

CSCI 8314: SPARSE MATRIX COMPUTATIONS

GENERAL INTRODUCTION

- *General introduction - a little history*
- *Motivation*
- *Resources*
- *What will this course cover*
- *Examples of problems leading to sparse matrix computations*

Historical Perspective: Focus of numerical linear algebra

➤ Linear algebra took many direction changes in the past

1940s–1950s: Major issue: flutter problem in aerospace engineering
→ eigenvalue problem [cf. Olga Taussky Todd] → LR, QR, .. → ‘EISPACK’

1960s: Problems related to the power grid promoted what we would call today general sparse matrix techniques

1970s– Automotive, Aerospace, ...: Computational Fluid Dynamics (CFD)

Late 1980s: Thrust on parallel matrix computations.

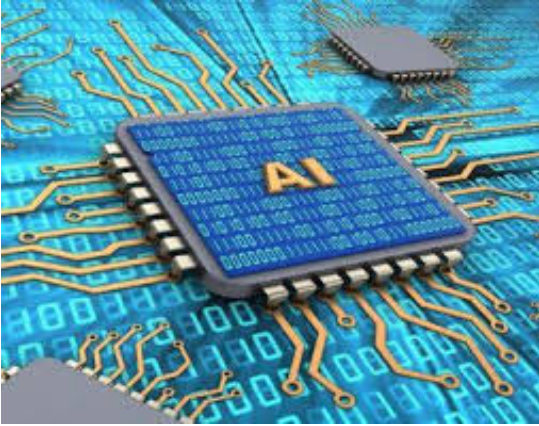
Late 1990s: Spur of interest in “financial computing”

Current: Machine Learning

Solution of PDEs (e.g., Fluid Dynamics) and problems in mechanical eng. (e.g. structures) major force behind numerical linear algebra algorithms in the past few decades.

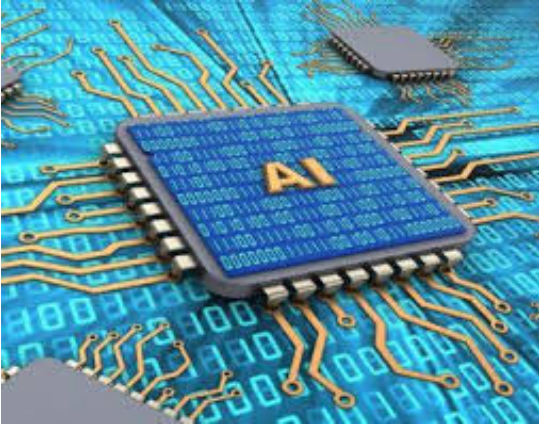
- Strong new forces are now reshaping the field today: Applications related to the use of “data”
- Machine learning is appearing in unexpected places:
 - design of materials
 - machine learning in geophysics
 - self-driving cars, ..
 -

Big impact on the economy



- New economy driven by Google, Facebook, Netflix, Amazon, Twitter, Ali-Baba, Tencent, ..., and even the big department stores (Walmart, ...)
- Huge impact on **Jobs**

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- Old leaders - e.g., Mining; Car companies; Aerospace; Manufacturing; offer little growth
 - Some instances of renewal driven by new technologies [e.g. Tesla]



- Look at what you are doing under new lenses: DATA

Sparse matrices: a brief history

Sparse matrices have been identified as important early on – terminology is quite old. Gauss defined the first method for such systems in 1823. Varga used explicitly the term 'sparse' in his 1962 book on iterative methods.

<https://www-users.cs.umn.edu/~saad/PDF/icerm2018.pdf>

- Special techniques used for sparse problems coming from Partial Differential Equations
- One has to wait until to the 1960s to see the birth of the general methodology available today
- Graphs introduced as tools for sparse Gaussian elimination in 1961 [Seymour Parter]

- Early work on reordering for banded systems, envelope methods
- Various reordering techniques for general sparse matrices introduced.
- Minimal degree ordering [Markowitz - 1957] ...
- ... later used in Harwell MA28 code [Duff] - released in 1977.
- Tinney-Walker Minimal degree ordering for power systems [1967]
- Nested Dissection [A. George, 1973]
- SPARSPAK [commercial code, Univ. Waterloo]
- Elimination trees, symbolic factorization, ...

History: development of iterative methods

- 1950s up to 1970s : focus on “relaxation” methods
- Development of ‘modern’ iterative methods took off in the mid-70s. but...
- ... The main ingredients were in place earlier [late 40s, early 50s: Lanczos; Arnoldi ; Hestenes (a local!) and Stiefel;]
- The next big advance was the push of ‘preconditioning’: in effect a way of combining iterative and (approximate) direct methods – [Meijerink and Van der Vorst, 1977]

<https://www-users.cs.umn.edu/~saad/PDF/NIST75th.pdf>

History: eigenvalue problems

- Another parallel branch was followed in sparse techniques for large eigenvalue problems.
- A big problem in 1950s and 1960s : flutter of airplane wings.. This leads to a large (sparse) eigenvalue problem
- Overlap between methods for linear systems and eigenvalue problems [Lanczos, Arnoldi]

Resources

- SuiteSparse site (Formerly : Florida collection)

<https://sparse.tamu.edu/>

- SPARSKIT, etc. [SPARSKIT = old written in Fortran. + more recent 'solvers']

<http://www.cs.umn.edu/~saad/software>

Resources – continued

Books: on sparse direct methods.

- Book by Tim Davis [SIAM, 2006] see syllabus for info
- An old reference [Still a great book]: Alan George and Joseph W-H Liu, **Computer Solution of Large Sparse Positive Definite Systems**, Prentice-Hall, 1981.
- Of interest, mostly for reference:
 - I. S. Duff and A. M. Erisman and J. K. Reid, **Direct Methods for Sparse Matrices**, Clarendon press, Oxford, 1986; 2nd edition: 2017.
 - Some coverage in Golub and van Loan [John Hopinks, 4th Ed., Chap. 10 to end]

BACKGROUND: PROBLEMS LEADING TO SPARSE MATRICES

Background: Examples leading to sparse matrices

- The classical: CFD, electrical networks,
- ... and the modern:
 - Graph algorithms and tools (Sparse graphs, graph coarsening, graphs and sparse methods). ..
 - Dimension reduction methods; Graph embeddings;
 - Specific machine learning algorithms; unsupervised/ supervised learning;
 - Deep learning;
 - Network analysis
 - ...

Example: Fluid flow

Physical Model



Nonlinear PDEs



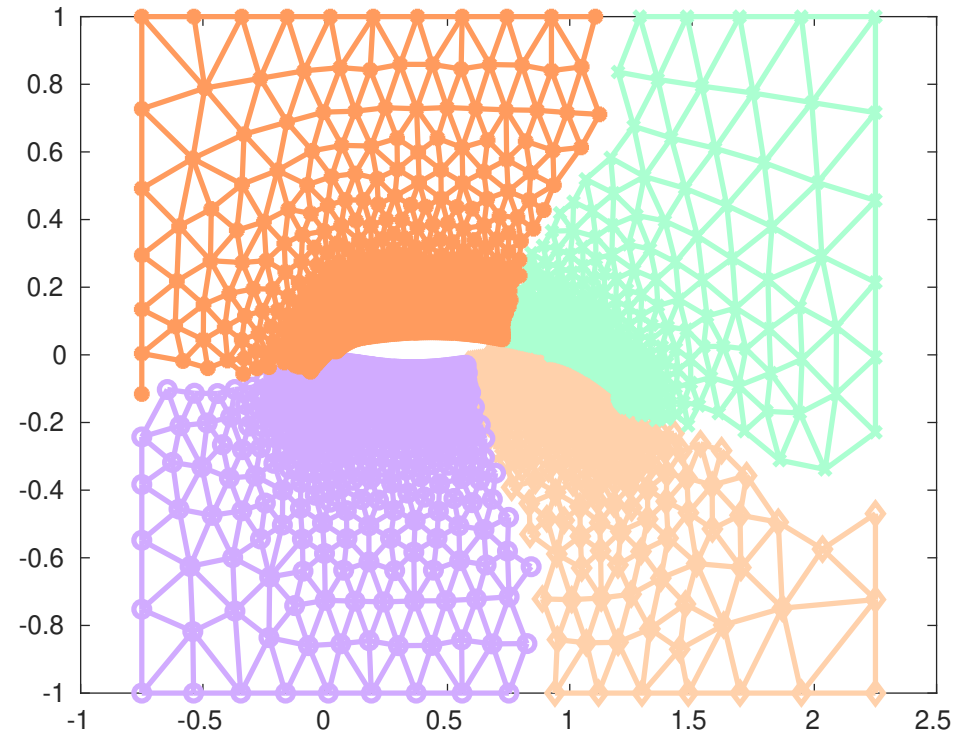
Discretization



Linearization (Newton)

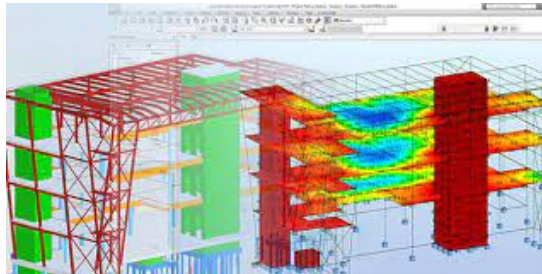
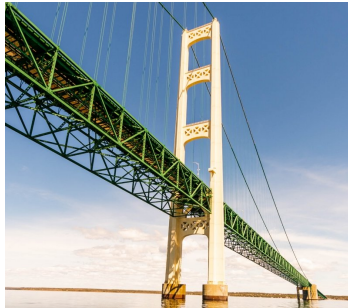
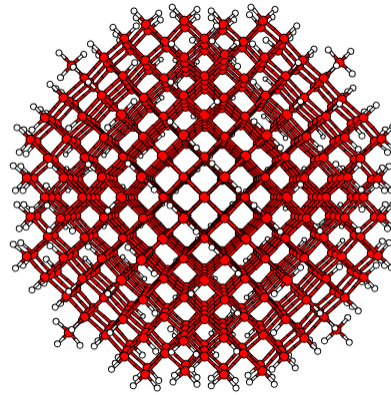
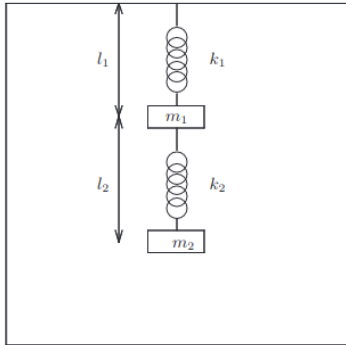


Sparse Linear Systems $Ax = b$



Example: Eigenvalue Problems

- Many applications require the computation of a few eigenvalues + associated eigenvectors of a matrix A

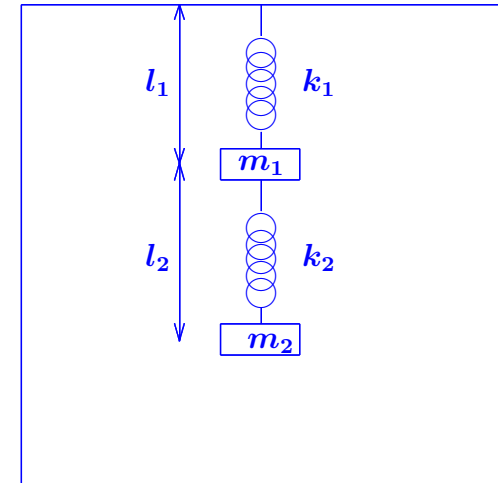


- Structural Engineering – (Goal: frequency response)
- Electronic structure calculations [Schrödinger equation..] – Quantum chemistry
- Stability analysis [e.g., electrical networks, mechanical system,..]
- ...

Example: Vibrations

- Vibrations in mechanical systems. See: www.cs.umn.edu/~saad/eig_book_2ndEd.pdf

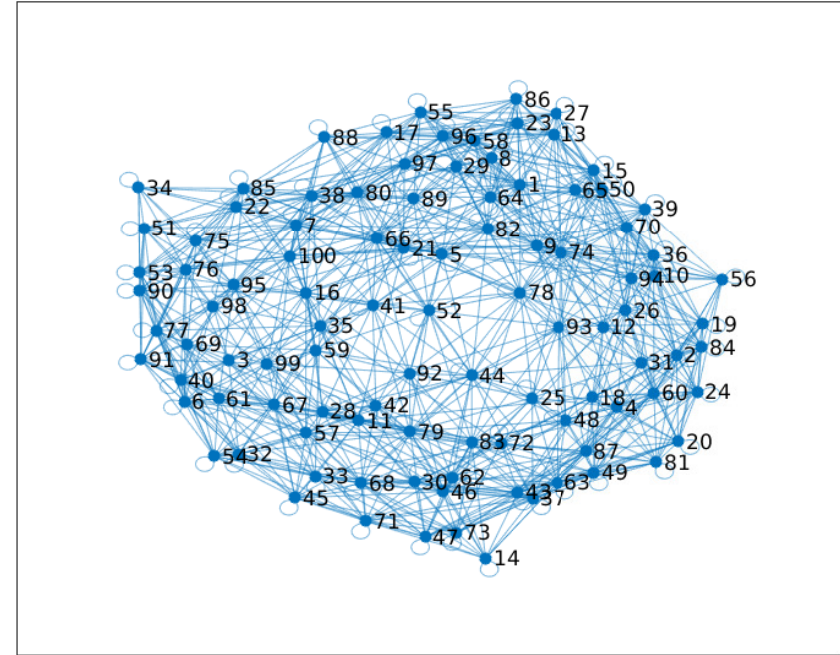
Problem: Determine the vibration modes of the mechanical system [to avoid resonance]. See details in Chapter 10 (sec. 10.2) of above reference.



- Problem type: Eigenvalue Problem

Example: Google Rank (pagerank)

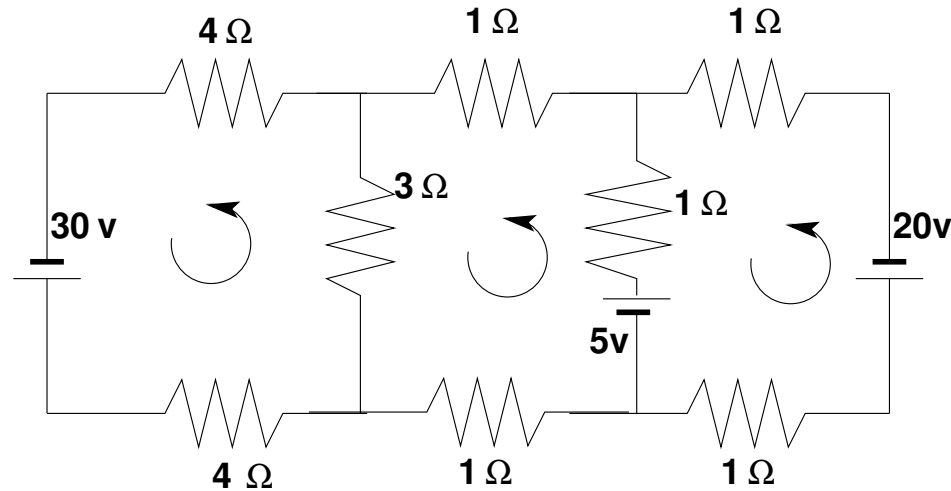
If one were to do a random walk from web page to web page, following each link on a given web page at random with equal likelihood, which are the pages to be encountered this way most often?



➤ Problem type: (homogeneous) Linear system. Eigenvector problem.

Example: Power networks

- Electrical circuits .. [Kirchhoff's voltage Law]



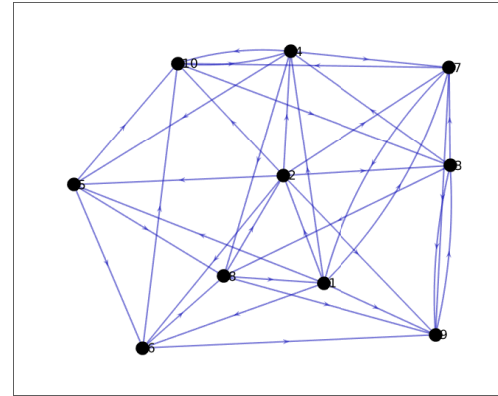
Problem: Determine the loop currents in a an electrical circuit - using Kirchhoff's Law ($V = RI$)

- Problem: Sparse Linear Systems [at the origin Sparse Direct Methods]

Example: Economics/ Marketing/ Social Networks

- Given: an influence graph G : g_{ij} = strength of influence of j over i
- Goal: charge member i price p_i in order to maximize profit
- Utility for member i : [x_i = consumption of i]

$$u_i = ax_i - bx_i^2 + \sum_{j \neq i} g_{ij}x_j - p_i x_i$$



- 1: 'Monopolist' fixes prices; 2: agent i fixes consumption x_i

Result: Optimal pricing proportional to **Bonacich** centrality:

$(I - \alpha G)^{-1} \mathbb{1}$ where $\alpha = \frac{1}{2b}$ [Candogan et al., 2012 + many refs.]

- 'centrality' defines a measure of importance of a node (or an edge) in a graph
- Many other ideas of centrality in graphs [degree centrality, betweenness centrality, closeness centrality, ...]
- Important application: Social Network Analysis

Example: Method of least-squares

- First use of least squares by Gauss, in early 1800's:

A planet follows an elliptical orbit according to $ay^2 + bxy + cx + dy + e = x^2$ in cartesian coordinates. Given a set of noisy observations of (x, y) positions, compute a, b, c, d, e , and use to predict future positions of the planet. This least squares problem is nearly rank-deficient and hence very sensitive to perturbations in the observations.

- Problem type: Least-Squares system

Read Wikipedia's article on planet ceres:

[http://en.wikipedia.org/wiki/Ceres_\(dwarf_planet\)](http://en.wikipedia.org/wiki/Ceres_(dwarf_planet))

Example: Dynamical systems and epidemiology

A set of variables that fill a vector y are governed by the equation

$$\frac{dy}{dt} = Ay$$

Determine $y(t)$ for $t > 0$, given $y(0)$ [called ‘orbit’ of y]

➤ Problem type: (Linear) system of ordinary differential equations.

Solution:

$$y(t) = e^{tA}y(0)$$

➤ Involves exponential of A [think Taylor series], i.e., a **matrix function**

- This is the simplest form of dynamical systems (linear).
- Consider the slightly more complex system:

$$\frac{dy}{dt} = A(y)y$$

- Nonlinear. Requires 'integration scheme'.

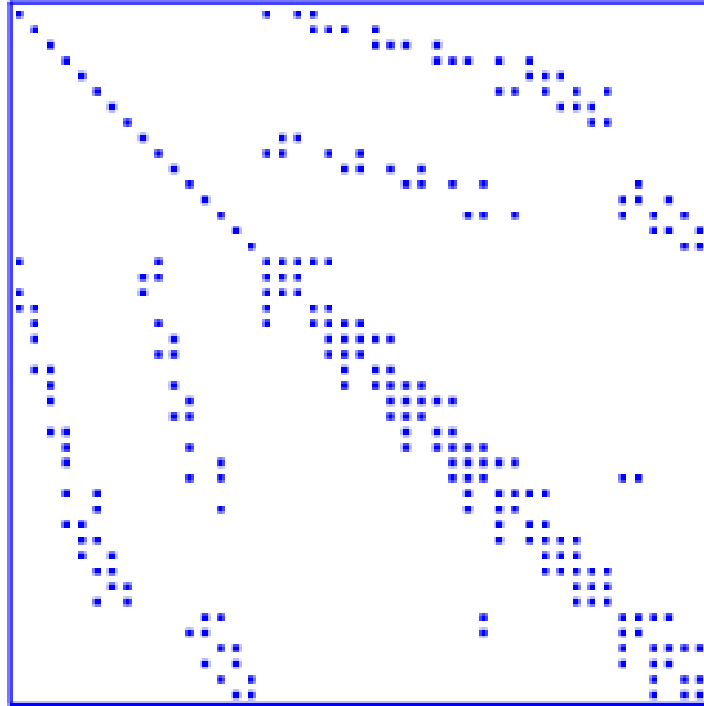
General Problems in Numerical Linear Algebra (dense & sparse)

- Linear systems: $Ax = b$. Often: A is large and sparse
- Least-squares problems $\min \|b - Ax\|_2$
- Eigenvalue problem $Ax = \lambda x$. Several variations -
- SVD .. and
- ... Low-rank approximation
- Tensors and low-rank tensor approximation
- Matrix equations: Sylvester, Lyapunov, Riccati, ..
- Nonlinear equations – acceleration methods
- Matrix functions and applications
- Many many more ...

SPARSE MATRICES

- *Introduction to sparse matrices*
- *See Chap. 3 of text*
- *See also the various sparse matrix sites*
- *Sparse matrices in matlab*

What are sparse matrices?



Pattern of a small sparse matrix

- Vague definition: matrix with few nonzero entries
- For all practical purposes: an $m \times n$ matrix is sparse if it has $O(\min(m, n))$ nonzero entries.
- This means roughly a constant number of nonzero entries per row and column -
- This definition excludes a large class of matrices that have $O(\log(n))$ nonzero entries per row.
- Other definitions use a slow growth of nonzero entries with respect to n or m .

“..matrices that allow special techniques to take advantage of the large number of zero elements.” (J. Wilkinson)

A few applications which lead to sparse matrices:

Structural Engineering, Computational Fluid Dynamics, Reservoir simulation, Electrical Networks, optimization, Google Page rank, information retrieval (LSI), circuit simulation, device simulation,

Goal of Sparse Matrix Techniques

- To perform standard matrix computations economically i.e., without storing the zeros of the matrix.

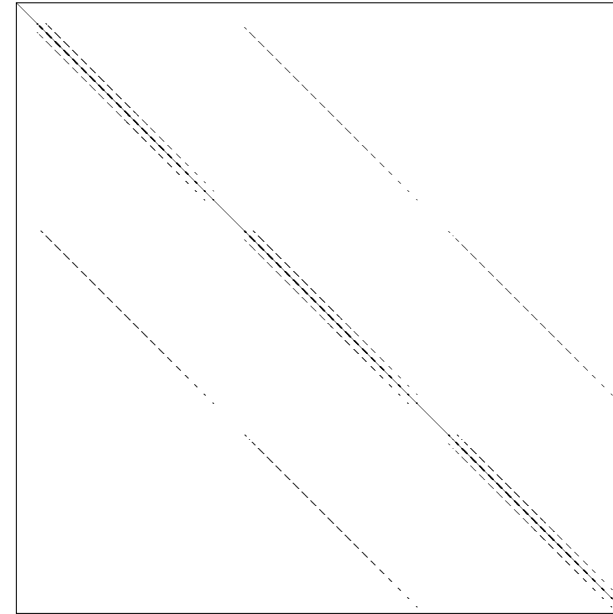
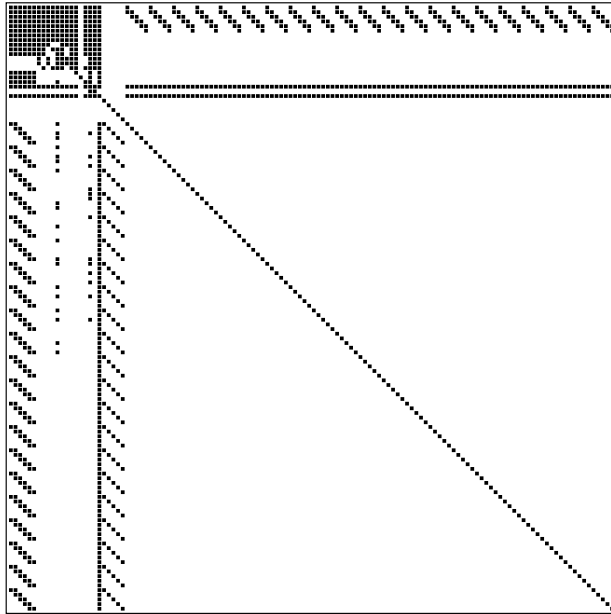
Example: To add two square dense matrices of size n requires $O(n^2)$ operations. To add two sparse matrices A and B requires $O(nnz(A) + nnz(B))$ where $nnz(X) =$ number of nonzero elements of a matrix X .

- For typical Finite Element /Finite difference matrices, number of nonzero elements is $O(n)$.

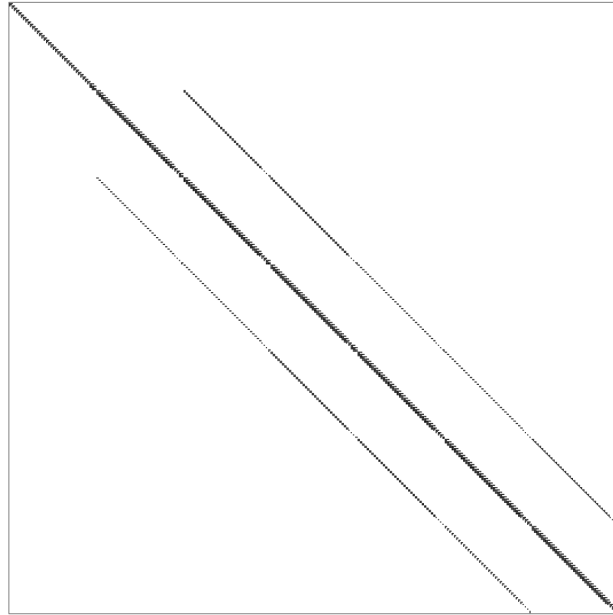
Remark: A^{-1} is usually dense, but L and U in the LU factorization may be reasonably sparse (if a good technique is used)

-  2 Look up Cayley-Hamilton's theorem if you do not know about it.
-  3 Show that the inverse of a matrix (when it exists) can be expressed as a polynomial of A , where the polynomial is of degree $\leq n - 1$.
-  4 When is the degree $< n - 1$? [Hint: look-up minimal polynomial of a matrix]
-  5 What is the pattern of the inverse of a tridiagonal matrix? a bidiagonal matrix?

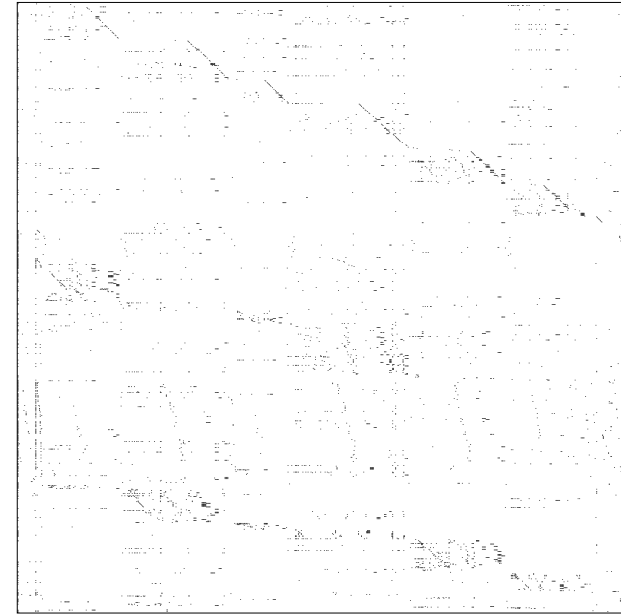
Nonzero patterns of a few sparse matrices



ARC130: Unsymmetric matrix from laser problem. a.r.curtis, oct 1974 SHERMAN5: fully implicit black oil simulator 16 by 23 by 3 grid, 3 unk



PORES3: Unsymmetric MATRIX FROM PORES

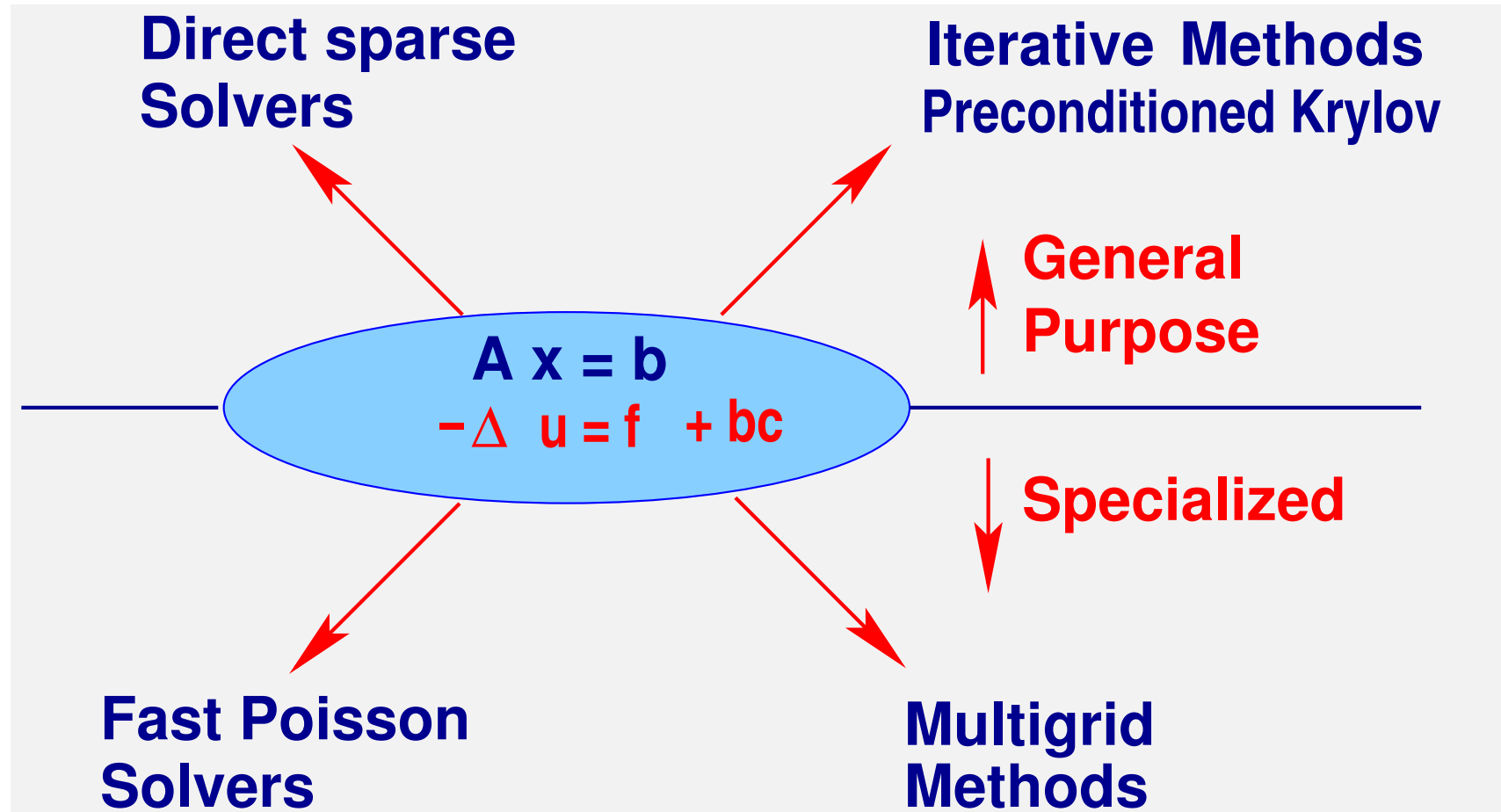


BP_1000: UNSYMMETRIC BASIS FROM LP PROBLEM BP

Types of sparse matrices

- Two types of matrices: structured (e.g. Sherman5) and **unstructured** (e.g. BP_1000)
- The matrices PORES3 and SHERMAN5 are from Oil Reservoir Simulation. Often: 3 unknowns per mesh point (Oil , Water saturations, Pressure). Structured matrices.
- 40 years ago reservoir simulators used rectangular grids.
- Modern simulators: Finer, more complex physics ➤ harder and larger systems. Also: unstructured matrices
- A naive but representative challenge problem: $100 \times 100 \times 100$ grid + about 10 unknowns per grid point ➤ $N \approx 10^7$, and $nnz \approx 7 \times 10^8$.

Solving sparse linear systems: existing methods



Two types of methods for general systems:

- Direct methods : based on sparse Gaussian elimination, sparse Cholesky,...
- Iterative methods: compute a sequence of iterates which converge to the solution - preconditioned Krylov methods..

Remark:

These two classes of methods have always been in competition.

- 40 years ago solving a system with $n = 10,000$ was a challenge
- Now you can solve this in a fraction of a second on a laptop.

- Sparse direct methods made huge gains in efficiency. As a result they are very competitive for 2-D problems.
- 3-D problems lead to more challenging systems [inherent to the underlying graph]

Difficulty:

- No robust 'black-box' iterative solvers.
 - At issue: Robustness in conflict with efficiency.
- Iterative methods are starting to use some of the tools of direct solvers to gain 'robustness'

Consensus:

1. Direct solvers are often preferred for two-dimensional problems (robust and not too expensive).
2. Direct methods loose ground to iterative techniques for three-dimensional problems, and problems with a large degree of freedom per grid point,

Sparse matrices in matlab

- Matlab supports sparse matrices to some extent.
- Can define sparse objects by conversion

```
A = sparse(X) ; X = full(A)
```

- Can show pattern

```
spy(X)
```

- Define the analogues of `ones`, `eye`:

```
speye(n,m) ,      spones(pattern)
```

- A few reorderings functions provided.. [will be studied in detail later]

```
symrcm, symamd, colamd, colperm
```

- Random sparse matrix generator:

```
sprand(S) or sprand(m,n, density)
```

(also `texttsprandn(...)`)

- Diagonal extractor-generator utility:

```
spdiags(A) , spdiags(B,d,m,n)
```

- Other important functions:

```
spalloc(..) , find(..)
```


Graph Representations of Sparse Matrices

- Graph theory is a fundamental tool in sparse matrix techniques.

DEFINITION. A graph G is defined as a pair of sets $G = (V, E)$ with $E \subset V \times V$. So G represents a binary relation. The graph is **undirected** if the binary relation is reflexive. It is **directed** otherwise. V is the vertex set and E is the edge set.

Example: Given the numbers 5, 3, 9, 15, 16, show the two graphs representing the relations

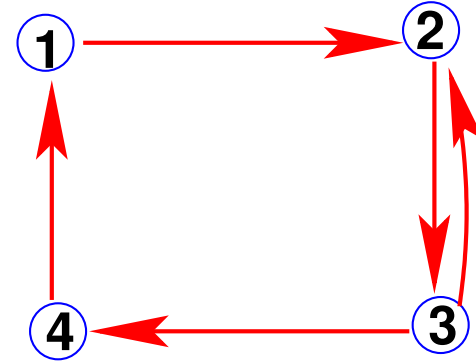
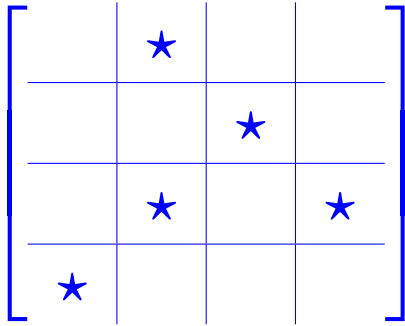
R1: Either $x < y$ or y divides x .

R2: x and y are congruent modulo 3. [$\text{mod}(x,3) = \text{mod}(y,3)$]

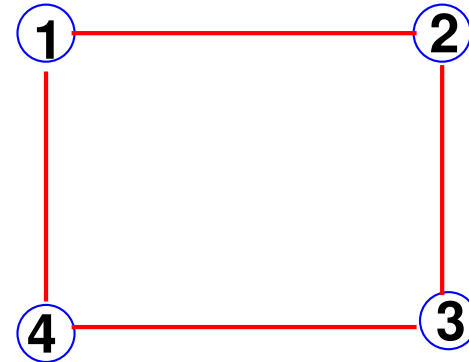
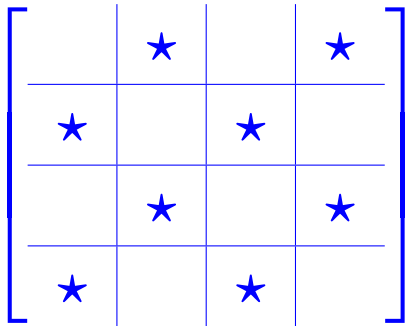
- Adjacency Graph $G = (V, E)$ of an $n \times n$ matrix A :
- Vertices $V = \{1, 2, \dots, n\}$.
 - Edges $E = \{(i, j) | a_{ij} \neq 0\}$.
- Often self-loops (i, i) are not represented [because they are always there]
- Graph is **undirected** if the matrix has a symmetric structure:

$$a_{ij} \neq 0 \quad \text{iff} \quad a_{ji} \neq 0.$$

Example: (directed graph)



Example: (undirected graph)



 6 Graph of a tridiagonal matrix? Of a dense matrix?

 7 Adjacency graph of:

$$A = \begin{bmatrix} \star & \star & & & \star & \\ \star & \star & \star & & & \star \\ & \star & \star & & & \\ & & & \star & \star & \\ \star & & & \star & \star & \star \\ & \star & & & \star & \star \end{bmatrix} ?$$

 8 Recall what a star graph is. Show a matrix whose graph is a star graph. Consider two situations: Case when center node is labeled first and case when it is labeled last.

- Note: Matlab now has a `graph` function.
- `G = graph(A)` creates adjacency graph from A
- G is a matlab class/
- `G.Nodes` will show the vertices of G
- `G.Edges` will show its edges.
- `plot(G)` will show a representation of the graph



Do the following:

- Load the matrix 'Bmat.mat' located in the class web-site (see 'matlab' folder)
- Visualize pattern (`spy(B)`) + find: Number of nonzero elements, size, ...
- Generate graph - without self-edges:

```
G = graph(B, 'OmitSelfLoops')
```

- Plot the graph –
- \$1M question: Any idea on how this plot is generated?

DISCRETIZATION OF PARTIAL DIFFERENTIAL EQUATIONS

Goal: to show how partial differential lead to sparse linear systems

- *See Chap. 2 of text*
- *Finite difference methods*
- *Finite elements*
- *Assembled and unassembled finite element matrices*

Why study discretized PDEs?

- One of the most important sources of sparse linear systems
- Will help understand the structures of the problem and their connections with “meshes” in 2-D or 3-D space
- Also: iterative methods are often formulated for the PDE directly – instead of a discretized (sparse) system.

NOTE: Useful to have an idea of how Finite Difference matrices are generated. For Finite Elements: goal is to unravel the related sparse computations to which they lead.

A typical numerical simulation

Physical Problem →

Nonlinear PDEs →

Discretization →

Linearization (Newton) →

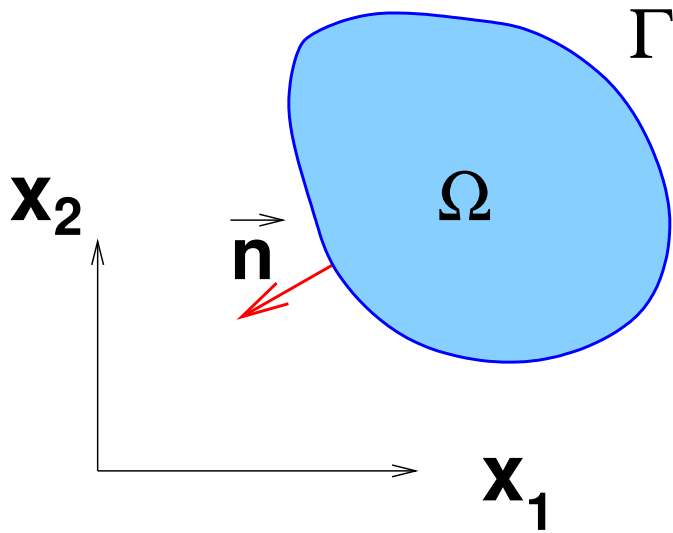
Sequence of Sparse Linear Systems $Ax = b$

Example: discretized Poisson equation

► Common Partial Differential Equation (PDE) :

$$\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} = f, \text{ for } x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ in } \Omega$$

where Ω = bounded, open domain in $\mathbb{R}^2$



► + boundary conditions:

Dirichlet: $u(x) = \phi(x)$

Neumann: $\frac{\partial u}{\partial \vec{n}}(x) = 0$

Cauchy: $\frac{\partial u}{\partial \vec{n}} + \alpha(x)u = \gamma$

- $\Delta = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}$ is the **Laplace** operator or **Laplacean**
- How to approximate the Poisson problem shown above?
- Answer: discretize, i.e., replace continuum with discrete set.
- Then approximate Laplacean using this discretization
- Many types of discretizations.. will briefly cover Finite Differences (FD) and Finite Elements (FEM)

Finite Differences: Basic approximations

- Formulas derived from Taylor series expansion:

$$u(x + h) = u(x) + h \frac{du}{dx} + \frac{h^2}{2} \frac{d^2u}{dx^2} + \frac{h^3}{6} \frac{d^3u}{dx^3} + \frac{h^4}{24} \frac{d^4u}{dx^4}(\xi)$$

Discretization of PDEs - Basic approximations

- Simplest scheme: forward difference

$$\begin{aligned}\frac{du}{dx} &= \frac{u(x+h) - u(x)}{h} - \frac{h}{2} \frac{d^2u(x)}{dx^2} + O(h^2) \\ &\approx \frac{u(x+h) - u(x)}{h}\end{aligned}$$

- Centered differences for second derivative:

$$\frac{d^2u(x)}{dx^2} = \frac{u(x+h) - 2u(x) + u(x-h))}{h^2} - \frac{h^2}{12} \frac{d^4u(\xi)}{dx^4},$$

where $\xi_- \leq \xi \leq \xi_+$.

Notation:

$$\delta^+ u(x) = u(x+h) - u(x)$$

$$\delta^- u(x) = u(x) - u(x-h)$$

➤ Operations of the type: $\frac{d}{dx} \left[a(x) \frac{d}{dx} \right]$ are common [in-homogeneous media].

➤ The following is a second order approximation:

$$\begin{aligned} \frac{d}{dx} \left[a(x) \frac{du}{dx} \right] &= \frac{1}{h^2} \delta^+ \left(a_{i-\frac{1}{2}} \delta^- u \right) + O(h^2) \\ &\approx \frac{a_{i+\frac{1}{2}}(u_{i+1} - u_i) - a_{i-\frac{1}{2}}(u_i - u_{i-1})}{h^2} \end{aligned}$$

 1 Show that $\delta^+ \left(a_{i-\frac{1}{2}} \delta^- u \right) = \delta^- \left(a_{i+\frac{1}{2}} \delta^+ u \right)$

Finite Differences for 2-D Problems

Consider the simple problem,

$$-\left(\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2}\right) = f \quad \text{in } \Omega \quad (1)$$

$$u = 0 \quad \text{on } \Gamma \quad (2)$$

$\Omega = \text{rectangle } (0, l_1) \times (0, l_2)$ and Γ its boundary.

Discretize uniformly :

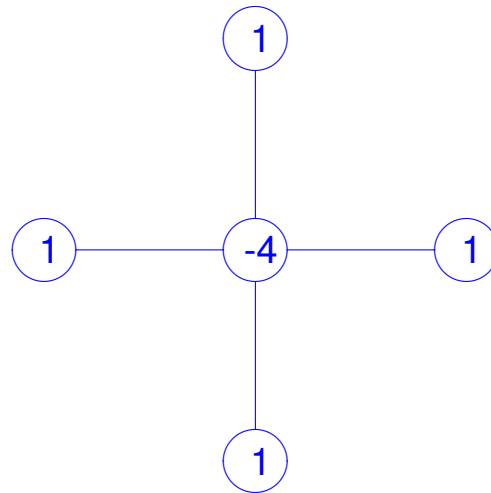
$$x_{1,i} = i \times h_1 \quad i = 0, \dots, n_1 + 1 \quad h_1 = \frac{l_1}{n_1 + 1}$$
$$x_{2,j} = j \times h_2 \quad j = 0, \dots, n_2 + 1 \quad h_2 = \frac{l_2}{n_2 + 1}$$

Finite Difference Scheme for the Laplacean

- Use centered differences for $\frac{\partial^2}{\partial x_1^2}$ and $\frac{\partial^2}{\partial x_2^2}$ - with mesh sizes $h_1 = h_2 = h$:

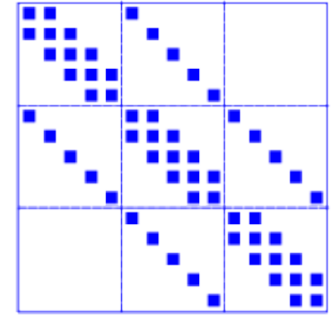
$$\Delta u(x) \approx \frac{1}{h^2} [u(x_1 + h, x_2) + u(x_1 - h, x_2) + u(x_1, x_2 + h) + u(x_1, x_2 - h) - 4u(x_1, x_2)]$$

The 5-point 'stencil:'



The resulting matrix has the following block structure:

$$A = \frac{1}{h^2} \begin{bmatrix} B & -I & \\ -I & B & -I \\ & -I & B \end{bmatrix} \rightarrow$$



Case: 7×5 grid

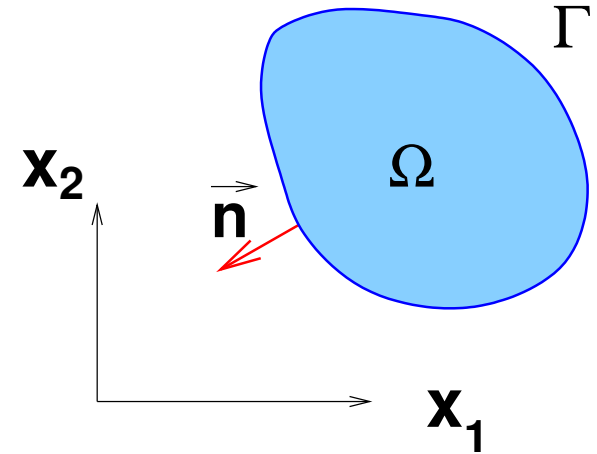
With

$$B = \begin{bmatrix} 4 & -1 & & & \\ -1 & 4 & -1 & & \\ & -1 & 4 & -1 & \\ & & -1 & 4 & -1 \\ & & & -1 & 4 \end{bmatrix}.$$

Finite Element Method (FEM): a quick overview

Background: Green's formula

$$\int_{\Omega} \nabla v \cdot \nabla u \, dx = - \int_{\Omega} v \Delta u \, dx + \int_{\Gamma} v \frac{\partial u}{\partial \vec{n}} \, ds.$$



➤ The dot indicates a dot product of two vectors.

➤ ∇ = gradient operator. In 2-D: $\nabla u = \begin{pmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \end{pmatrix},$

➤ Δu = Laplacean of u

➤ $\vec{n}$ is the unit vector that is normal to Γ and directed outwards.

➤ Frechet derivative:

$$\frac{\partial u}{\partial \vec{v}}(x) = \lim_{h \rightarrow 0} \frac{u(x + h\vec{v}) - u(x)}{h}$$

➤ Green's formula generalizes the usual formula for integration by parts

➤ Define

$$a(u, v) \equiv \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} \left(\frac{\partial u}{\partial x_1} \frac{\partial v}{\partial x_1} + \frac{\partial u}{\partial x_2} \frac{\partial v}{\partial x_2} \right) dx$$
$$(f, v) \equiv \int_{\Omega} f v \, dx.$$

➤ With Dirichlet BC, the integral on the boundary in Green's formula vanishes →

$$a(u, v) = -(\Delta u, v).$$

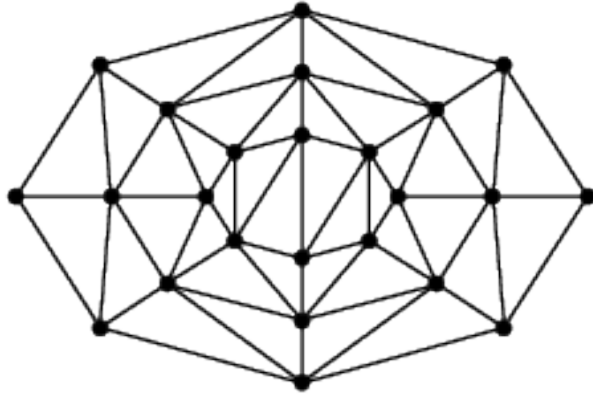
- Suppose we want to solve $-\Delta u = f$ in Ω + *Dirichlet BC*
- **Weak formulation** of the original problem: select a subspace of reference V of L^2 and then solve

$$\text{Find } u \in V \text{ such that } \underbrace{a(u, v)}_{=-(\Delta u, v)} = (f, v), \quad \forall v \in V$$

- Finite Element method solves this weak problem...
- ... by discretization

- The original domain is approximated by the union Ω_h of m triangles K_i ,

Triangulation of Ω :



$$\Omega_h = \bigcup_{i=1}^m K_i.$$

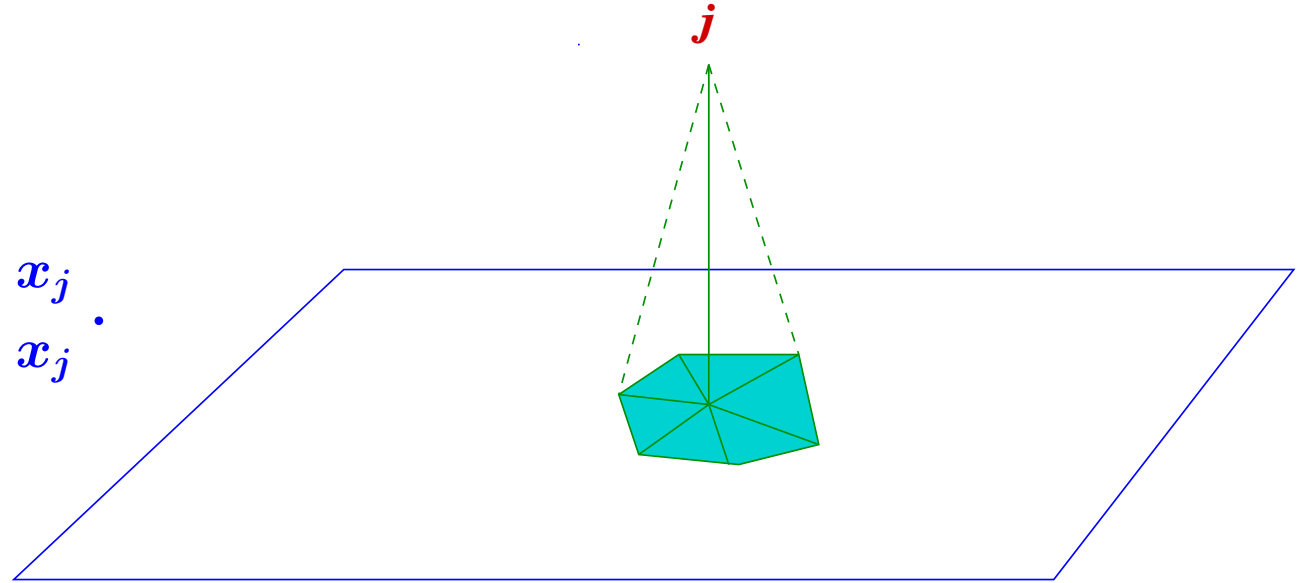
- Some restrictions on angles, edges, etc..

$$V_h = \{ \phi \mid \phi|_{\Omega_h} \in \mathcal{C}^0, \quad \phi|_{\Gamma_h} = 0, \quad \phi|_{K_j} \text{ linear } \forall j \}$$

- $\mathcal{C}^0$ = set of *continuous* functions
- $\phi|_X$ == restriction of ϕ to the subset X
- Let $x_j, j = 1, \dots, n$, be the nodes of the triangulation

- Can define a (unique) 'hat' function ϕ_j in V_h associated with each x_j s.t.:

$$\phi_j(x_i) = \delta_{ij} = \begin{cases} 1 & \text{if } x_i = x_j \\ 0 & \text{if } x_i \neq x_j \end{cases}.$$



- Each function u of V_h can be expressed as

$$u(x) = \sum_{j=1}^n \xi_j \phi_j(x). \quad (*)$$

- FEM approximation $\equiv$ Galerkin condition for functions in V_h :

Find $u \in V_h$ such that $a(u, v) = (f, v), \quad \forall v \in V_h$

- Express u in the basis $\{\phi_j\}$ (see *), then substitute above. Result:

- Linear system $\sum_{j=1}^n \alpha_{ij} \xi_j = \beta_i$ where: $\alpha_{ij} = a(\phi_j, \phi_i), \quad \beta_i = (f, \phi_i).$

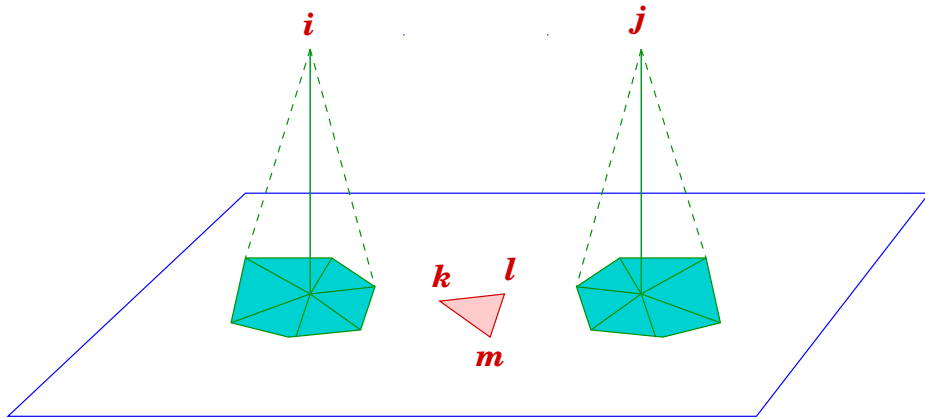
The above equations form a linear system of equations

$$Ax = b$$

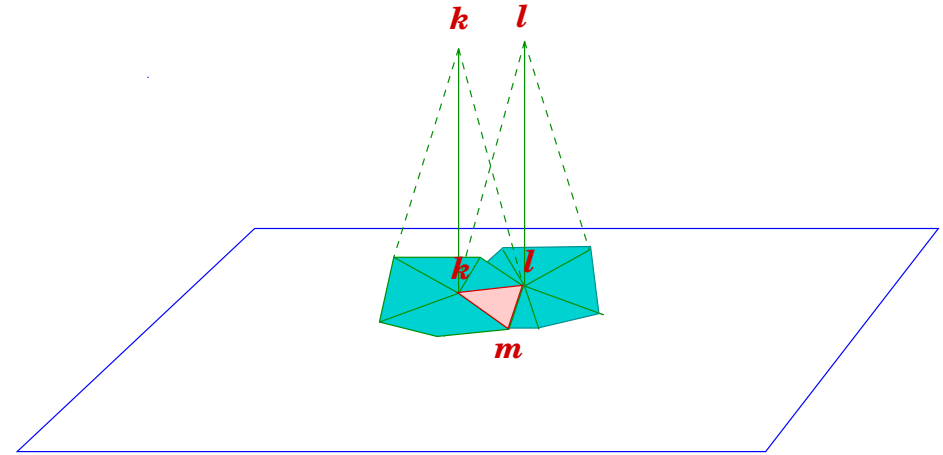
- A is *Symmetric Positive Definite*

 2 Prove it

The Assembly Process: Illustration



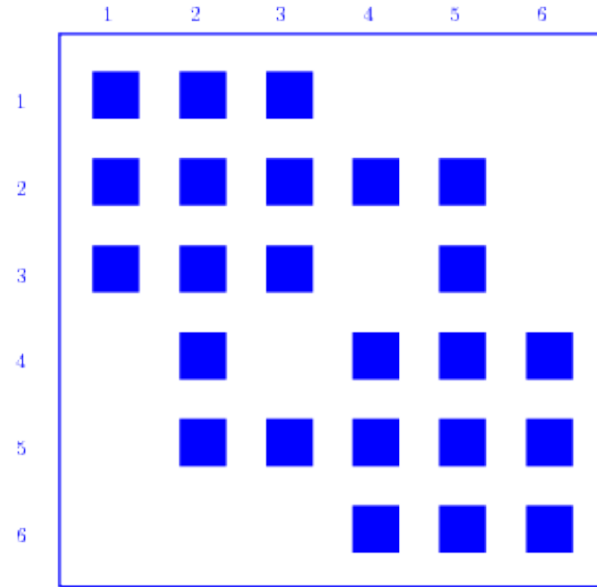
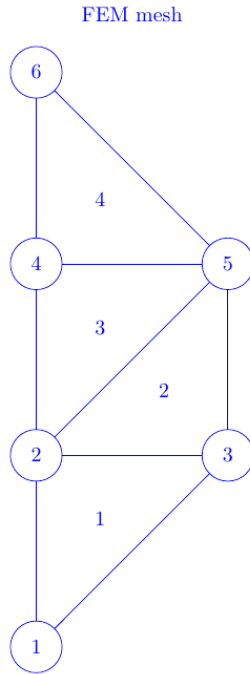
If triangle $K \notin$ support domains of both ϕ_i and ϕ_j then $a_K(\phi_i, \phi_j) = 0$



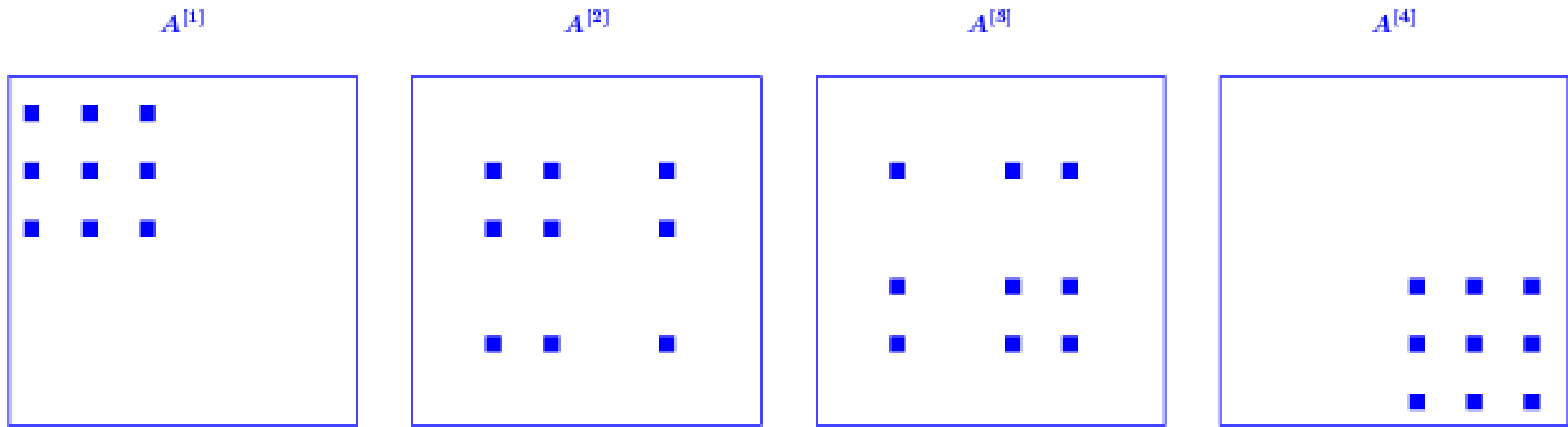
If triangle $K \in$ *both* nonzero domains of ϕ_i and ϕ_j then $a_K(\phi_i, \phi_j) \neq 0$

► So: $a_K(\phi_i, \phi_j) \neq 0$ iff $i \in \{k, l, m\}$ and $j \in \{k, l, m\}$.

The Assembly Process



Small finite element mesh and pattern of the corresponding assembled matrix



Element matrices $A^{[e]}$, $e = 1, \dots, 4$ for FEM mesh shown above

- Each element contributes a 3×3 submatrix $A^{[e]}$ (spread out)
- Can also use the matrix in un-assembled form - To multiply a vector by A for example we can do:

$$y = Ax = \sum_{e=1}^{nel} A^{[e]}x = \sum_{e=1}^{nel} P_e A_{K_e} (P_e^T x)$$

- Can be computed using the element matrices A_{K_e} - no need to assemble
- The product $P_e^T x$ gathers x data associated with the e -element into a 3-vector consistent with the ordering of the matrix A_{K_e} .
- Advantage: some simplification in process
- Disadvantage: cost (memory + computations).

Resources: A few matlab scripts

- These (and others) will be posted in the matlab folder of class web-site

```
>> help fd3d
```

```
function A = fd3d(nx,ny,nz,alpx,alpy,alpz,dshift)
```

```
NOTE nx and ny must be > 1 -- nz can be == 1.
```

```
5- or 7-point block-Diffusion/conv. matrix. with
```

- A stripped-down version is `lap2D(nx,ny)`

```
>> help mark
```

```
[A] = mark(m)
```

```
generates a Markov chain matrix for a random walk
```

```
on a triangular grid. A is sparse of size  $n=m*(m+1)/2$ 
```

3 Explore A few useful matlab functions

- * `kron`
- * `gplot` for plotting graphs
- * `reshape` for going from say 1-D to 2-D or 3-D arrays

 4 Write a script to generate a 9-point discretization of the Laplacean.

